

Derivative Rules

- Addition Rule: $\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$
- Multiplicative Constant Rule: $\frac{d}{dx}(k \cdot u(x)) = k \cdot \frac{du}{dx}$
- Product Rule: $\frac{d}{dx}(u(x) \cdot v(x)) = \frac{du}{dx} \cdot v(x) + \frac{dv}{dx} \cdot u(x)$
- Quotient Rule: $\frac{d}{dx}\left(\frac{u(x)}{v(x)}\right) = \frac{\frac{du}{dx} \cdot v(x) - \frac{dv}{dx} \cdot u(x)}{v(x)^2}$
- Power Rule: $\frac{d}{dx}(x^k) = k \cdot x^{k-1}$
- Chain Rule: $\frac{d}{dx}(u(v(x))) = \frac{d}{dv} u(v(x)) \cdot \frac{d}{dx} v(x) = u'(v(x)) \cdot v'(x)$

Derivatives for Special Functions

- $\frac{d}{dx} e^x = e^x$
- $\frac{d}{dx} \ln x = \frac{1}{x}$
- $\frac{d}{dx} \sin x = \cos x$
- $\frac{d}{dx} \cos x = -\sin x$

Deriving the Addition Rule

- $\frac{d}{dx}(u(x) + v(x)) = \lim_{h \rightarrow 0} \frac{(u(x+h) + v(x+h)) - (u(x) + v(x))}{h}$
- $= \lim_{h \rightarrow 0} \frac{(u(x+h) - u(x)) + (v(x+h) - v(x))}{h}$
- $= \lim_{h \rightarrow 0} \left(\frac{u(x+h) - u(x)}{h} + \frac{v(x+h) - v(x)}{h} \right)$
- $\frac{df}{dx} + \frac{dg}{dx}$

Deriving the Multiplicative Constant Rule

- $\frac{d}{dx}(k \cdot u(x)) = \lim_{h \rightarrow 0} \frac{k(u(x+h)) - k(u(x))}{h}$
- $= \lim_{h \rightarrow 0} k \frac{u(x+h) - u(x)}{h}$
- $= k \cdot \frac{du}{dx}$

Example of Addition and Constant Rules

Consider the functions of $j(x) = 2x^2 + 4x$, $f(x) = x^2$, and $g(x) = x$.

We will show that $\frac{dj(x)}{dx} = 2\frac{df(x)}{dx} + 4\frac{dg(x)}{dx}$

$$\begin{aligned} \blacksquare \quad \frac{dj(x)}{dx} &= \lim_{h \rightarrow 0} \frac{(2(x+h)^2 + 4(x+h)) - (2x^2 + 4x)}{h} \\ \blacksquare &= \lim_{h \rightarrow 0} \frac{2(x+h)^2 - 2x^2 + 4(x+h) - 4x}{h} \\ \blacksquare &= \lim_{h \rightarrow 0} 2 \frac{(x+h)^2 - x^2}{h} + \lim_{h \rightarrow 0} 4 \frac{(x+h) - x}{h} \\ \blacksquare &= 2 \frac{df(x)}{dx} + 4 \frac{dg(x)}{dx} \end{aligned}$$

Deriving the Chain Rule

Let f and g be differentiable functions.

- $\frac{dg(f(x))}{dx} = \lim_{h \rightarrow 0} \frac{g(f(x+h)) - g(f(x))}{h}$
- Multiply and divide by $f(x+h) - f(x)$
- $= \lim_{h \rightarrow 0} \frac{g(f(x+h)) - g(f(x)) (f(x+h) - f(x))}{h(f(x+h) - f(x))}$
- $= \lim_{h \rightarrow 0} \frac{g(f(x+h)) - g(f(x))}{f(x+h) - f(x)} \frac{f(x+h) - f(x)}{h}$
- Since f is differentiable it is also continuous. This means that $f(x+h) - f(x) \rightarrow 0$ as $h \rightarrow 0$.
- Let $z = f(x+h) - f(x)$ and $u = f(x)$. So $g(f(x+h)) = g(u+z)$
- $= \lim_{z \rightarrow 0} \frac{g(u+z) - g(u)}{z} \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- $= \frac{dg(u)}{du} \frac{du}{dx}$, where $u = f(x)$

Example of the Chain Rule

Consider the function $h(x) = (3x + 6)^2$.

- Break this function into two functions: $g(x) = x^2$ and $f(x) = 3x + 6$.
- Apply the chain rule: $\frac{dg(u)}{du} \frac{df(x)}{dx} = 2(3x + 6)3 = 6(3x + 6)$

Partial Statement of the Inverse Function Theorem

We can use the chain rule to compute the derivative of the inverse of a function. Recall that for an inverse of a function

$f(f^{-1}(x)) = x$. Using the chain rule we get:

- $\frac{df(f^{-1}(x))}{dx} = \frac{df(u)}{du} \frac{df^{-1}(x)}{dx}$, where $u = f^{-1}(x)$
- $\frac{dx}{dx} = f'(f^{-1}(x)) \frac{df^{-1}(x)}{dx}$
- $\frac{df^{-1}(x)}{dx} = \frac{1}{f'(f^{-1}(x))}$
- Consider $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$, $f(x) = x^2$ and $f^{-1}(x)$, then $f'(x) = 2x$ and $\frac{df^{-1}(x)}{dx} = \frac{1}{f'(f^{-1}(x))} = \frac{1}{2x^{1/2}}$

Deriving Product Rule

- $\frac{d(f(x)g(x))}{dx} = (fg)' = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}$
- Add and Subtract the term $f(x)g(x+h)$ from the RHS
- $(fg)' = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h}$
- $= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h)}{h} + \lim_{h \rightarrow 0} \frac{f(x)g(x+h) - f(x)g(x)}{h}$
- $= (\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h})(\lim_{h \rightarrow 0} g(x+h)) +$
 $(\lim_{h \rightarrow 0} f(x))(\lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h})$ (Pulling out Common Terms)
- $= f'g + fg'$

Product Rule Examples

- $f(x) = (x + 1)(x^2 + 3)$. What is $f'(x)$?
 - $f'(x) = 1(x^2 + 3) + 2x(x + 1)$
- $f(x) = \frac{x+1}{(x+2)^2}$
- Rewrite Function $f(x) = (x + 1)(x + 2)^{-2}$
- Apply Product and Chain Rule:
 $\frac{df(x)}{dx} = (x + 2)^{-2} - 2(x + 1)(x + 2)^{-3}$

Deriving the Power Rule

$$\blacksquare \quad \frac{dx^n}{dx} = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} = \lim_{h \rightarrow 0} \frac{x^n + nhx^{n-1} - x^n}{h} = nx^{n-1}$$

Applying the Power Rule

- $f(x) = x^8$
- $f(x) = 10$
- $f(x) = \frac{1}{x^4}$
- $f(x) = x^{1/3}$

Deriving the derivative of e^x

- $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} \dots$
- $\frac{de^x}{dx} = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = \lim_{h \rightarrow 0} \frac{e^x e^h - e^x}{h} = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h}$
- $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = \lim_{h \rightarrow 0} \frac{-1 + 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \dots}{h} = 1$

General Exponential Rule

- $a^x = e^{\ln(a^x)} = e^{x \ln(a)}$
- Let $g = e^x$ and $f = x \ln(a)$ and note that $\frac{dg}{dx} = e^x$ and $\frac{df}{dx} = \ln(a)$
- $a^x = g(f(x))$ and $\frac{da^x}{dx} = g'(f(x))f'(x)$
- $\frac{da^x}{dx} = e^{x \ln(a)}(\ln(a))$
- $\frac{da^x}{dx} = a^x(\ln(a))$

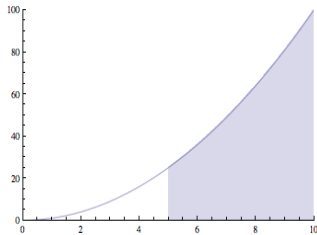
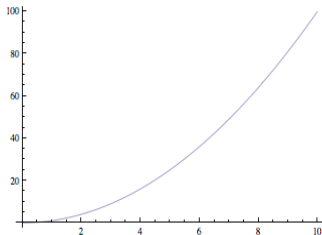
Derivatives of Logarithms

Recall that $e^{\ln x} = x$. The Logarithm is the inverse function of the exponential function. So we can apply the inverse function rule.

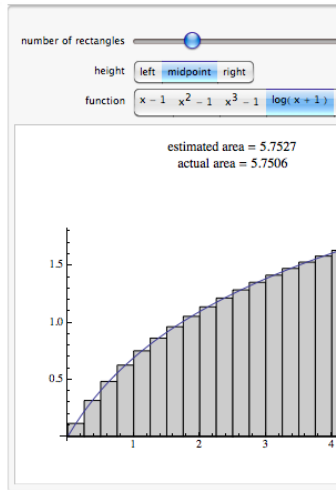
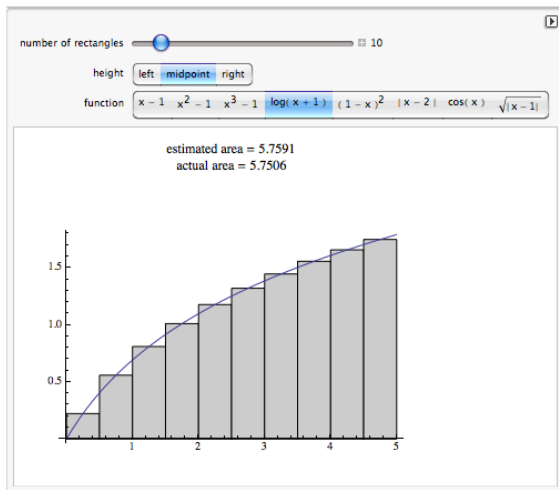
- $\frac{df^{-1}(x)}{dx} = \frac{1}{f'(f^{-1}(x))}$
- Define $f(x) = e^x$ and $f^{-1}(x) = \ln(x)$ and note that $f'(x) = e^x$
- $\frac{d \ln(x)}{dx} = \frac{1}{e^{\ln x}} = \frac{1}{x}$
- $\frac{d \log_a(x)}{dx} = \frac{1}{x \ln(a)}$
- Let $y = \ln(f(x))$, then $y' = \frac{f'(x)}{f(x)}$, which provides the relative rate of change.

Definite Integrals

Consider the function $f(x) = x^2$. A definite integral, calculated, for example, between 5 and 10, allows us to calculate the area under the graph between points $x = 5$ and $x = 10$. This is expressed as $\int_5^{10} f(x) dx$.



Riemann Sums



Indefinite Integrals

If $f(x)$ is given then any function $F(x)$ such that $F'(x) = f(x)$ is called an indefinite integral of $f(x)$, or the anti-derivative. Note that there are infinitely many anti-derivatives of a function $f(x)$ since they can differ by a constant. We denote anti-derivatives by $\int f(x)dx$. Rules for finding anti-derivatives:

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$
- $\int e^x dx = e^x + C$
- $\int a^x \ln(a) dx = a^x + C$
- $\int f'(x) e^{f(x)} dx = e^{f(x)} + C$
- $\int \frac{f'(x)}{f(x)} dx = \ln[f(x)] + C$

The Fundamental Theorem of Calculus

If $f(x)$ is continuous on $[a, b]$, and $F(x)$ is the anti-derivative of $f(x)$, then $\int_a^b f(x)dx = F(b) - F(a)$. The integral of f from a to b is nothing else than the anti-derivative, F , computed at b minus the anti-derivative, F , at a .

The Integral as a Linear Operator

$$\int [af(t) + bg(t)] dt = a \int f(t) dt + b \int g(t) dt$$

- Define the anti-derivatives $ay(x) = a \int \frac{dy}{dx} dx$ and $bz(x) = b \int \frac{dz}{dx} dx$
- Adding the anti-derivatives we get $ay + bz = a \int \frac{dy}{dx} dx + b \int \frac{dz}{dx} dx$
- Using the fact that the derivative is a linear operator:
$$\frac{d(ay+bz)}{dx} = a \frac{dy}{dx} + b \frac{dz}{dx}$$
- Integrate both sides: $\int \frac{d(ay+bz)}{dx} dx = \int (a \frac{dy}{dx} + b \frac{dz}{dx}) dx$
- Using the definition of the anti-derivative:
$$ay + bz = \int (a \frac{dy}{dx} + b \frac{dz}{dx}) dx$$
- These statements imply that
$$\int (a \frac{dy}{dx} + b \frac{dz}{dx}) dx = a \int \frac{dy}{dx} dx + b \int \frac{dz}{dx} dx$$

Properties of Integrals

- $\int_a^b [f(t) + g(t)] dt = \int_a^b f(t) dt + \int_a^b g(t) dt$
- $\int_a^b \lambda f(t) dt = \lambda \int_a^b f(t) dt$
- $\int_a^c f(t) dt = \int_a^b f(t) dt + \int_b^c f(t) dt$
- $\int_a^b f(t) dt = - \int_b^a f(t) dt$
- $\int_a^a f(t) dt = 0$

Integration by Substitution

To find an integral of the form $\int_a^b f(g(x))g'(x)dx$ one can apply the substitution rule:

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$

where $u = g(x)$ and $du = g'(x)dx$.

Integration by Substitution: Example

Find $\int (x+1)^{10} dx$.

For this example $f(x) = x^{10}$, $g(x) = x+1$, and $g'(x) = 1$. Let $u = x+1$. This implies $du = dx$.

Substituting this into the equation above we know have the integral $\int u^{10} du = \frac{u^{11}}{11}$.

Substituting this back in for u we find that $\int (x+1)^{10} dx = \frac{(x+1)^{11}}{11}$. This is easier than expanding the function and finding the integral.

Example 2

Consider the integral $\int (\frac{1}{2\pi} x e^{\frac{x^2}{2}}) dx$, the expected value of the the standard normal multiplied by x .

- Set $u = g(x) = -\frac{x^2}{2}$ and note that $g'(x) = -x$
- Then $\int \{ \frac{-1}{2\pi} g'(x) e^{g(x)} dx = \int (\frac{-1}{2\pi} e^u) du$
- $= \frac{-1}{2\pi} e^u + C$
- $= \frac{-1}{2\pi} e^{-\frac{x^2}{2}} + C$

Integration by Parts

To find an integral of the form $\int_a^b f(x) \cdot g'(x) dx$ one can apply the integration by parts rule:

$$\int f(x) \cdot g'(x) dx = f(x) \cdot g(x) - \int g(x) \cdot f'(x) dx$$

Integration by Parts: Example

Find $\int \ln(x) dx$.

Let $f(x) = \ln(x)$ and $g'(x) = 1$. Then $f'(x) = \frac{1}{x} dx$ and $g(x) = x$. So

$$\begin{aligned}\int f(x) \cdot g'(x) dx &= f(x) \cdot g(x) - \int g(x) \cdot f'(x) dx = \\ &= x \ln(x) - \int x \cdot \frac{1}{x} dx = x[\ln(x) - 1]\end{aligned}$$

Integration by Parts: Example 2

Find $\int xe^x dx$

- How can we define xe^x in terms of $f(x)$ and $g'(x)$?
- $f(x) = x$ and $g'(x) = e^x$.
- Then we can redefine the integral as
$$xe^x - \int e^x dx = e^x(x - 1) + C$$